

# **Tool Code Library System**

## **Official User Manual**

A guide for Requesters, Approvers, and System Administrators.

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## 1 System Overview & Basics

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The Tool Code Library System is a centralised platform designed to standardise your tool management workflow. It ensures that unique Tool Codes are generated, reviewed, and finalised accurately through a structured approval process, whether it is a one-time entry or a complex multi-supplier specification.

### Key Features

- Automated, sequential Tool Code generation (maximum 40 characters) with guaranteed uniqueness.
- Role-Based Access Control (RBAC) with four roles: Requester, Approver, Administrator, and Super Admin.
- Structured two-step approval workflow with documented rejection reasons.
- Master Data management for Suppliers, Materials, Joint Types, and dynamic Field Attributes.
- Real-time reporting with date-range filters and instant keyword search.

### Best Practices

- **Supported Browsers:** For the best experience, use the latest versions of Chrome, Edge, Safari, or Firefox.
- **Supplier Selection:** When filling a Tool Code request, Supplier 1 and Supplier 2 must always be different. The system will block submission if they are identical.
- **Drafts:** Use *Save as Draft* whenever your information is incomplete. You can return and finish the form at any time without losing progress.

## 2 Accessing the System

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### Logging In

1. Navigate to the Tool Code Library System login page.
2. Enter your assigned **Username** and **Password**.
3. Click **Login**. (*Note: If you forget your password, please contact a System Administrator to reset it.*)

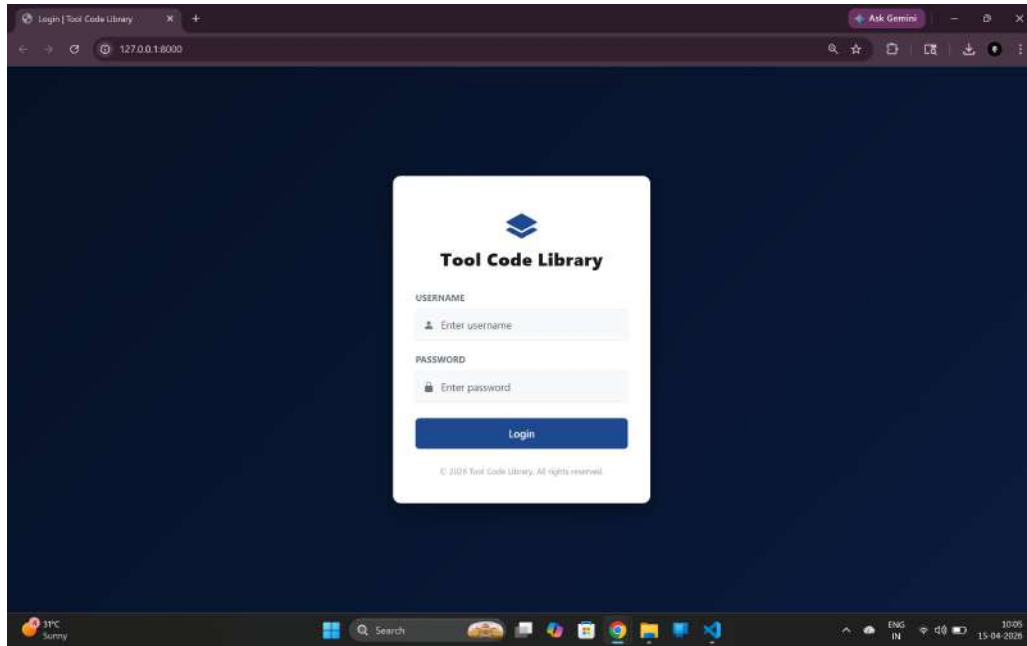


Figure 1: The Login Page Interface

### Viewing Your Profile

Once logged in, you can view your account details by clicking on your username in the navigation bar. Here you will see your **Account Type** (User, Admin, or Super Admin), the email associated with your account, and your last login date.

If your account has been deactivated by an Administrator, the system will block authentication and display an error message on the login screen.

## 3 Navigating the Dashboard

The Dashboard is your primary workspace after login. It is divided into three main sections:

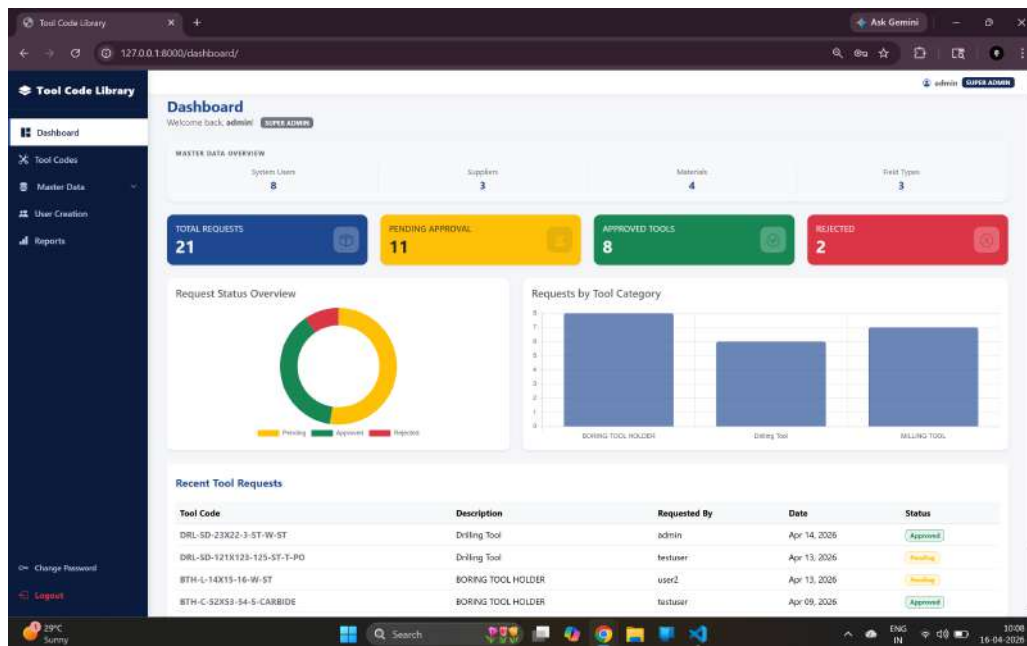


Figure 2: The Main Dashboard View

## 1. The Summary Bar

At the top of the screen, you will see three quick-glance metrics:

- **Total Requests:** The total number of Tool Code requests across all statuses in your account.
- **Pending Approval:** Requests that have been submitted and are awaiting an Approver's decision.
- **Approved Tools:** Requests that have been fully approved and whose Tool Codes are active in the system.

## 2. Analytics Charts

- **Request Status Overview (Donut Graph):** Displays the proportional breakdown of all request statuses (Draft, Pending, Approved, Rejected).
- **Requests by Tool Category (Bar Graph):** Shows how requests are distributed across Primary Description types (e.g., Drill, Mill).

## 3. Recent Tool Requests Table

This table provides a read-only overview of the five most recently submitted Tool Code requests.

- **Quick Visibility:** Instantly check the Tool Code, Primary Description, Requester, Submission Date, and current Status (Pending, Approved, Rejected, or Draft).

- **Action Alerts (Approvers Only):** If you have approval permissions, a **yellow Action Required banner** will appear at the top of the dashboard indicating the number of requests awaiting your urgent review.

## 4 Creating & Managing Tool Code Requests

### Step 1: Basic Information

To create a new request, navigate to the **Tool Codes** tab via the sidebar. Click the **+ Create Code** button located in the top right corner above the tool requests table.

- **Primary Description (Required):** Select the top-level tool category (e.g., Drill, Mill). The form will *dynamically expand* to show the specific sub-fields required for that tool type.
- **Dynamic Sub-Fields:** Complete all tool-specific attributes that appear, such as Flute Length or Bore Size. Required fields are marked with an asterisk (\*).
- **Supplier 1 & Supplier 2:** Select both suppliers from the dropdowns. These two selections must be different.
- **Joint Type:** Select the applicable joint type classification.
- **Raw Material:** Select the material for this tool specification.
- **Remarks (Optional):** Add any internal notes or context for the Approver.

Figure 3: Create Code Form — dynamic sub-fields expanded after Primary Description selection

### Step 2: Saving and Submitting

After completing the form, you have two options:

- **Save as Draft:** If information is missing, click **Save as Draft**. This bypasses strict validation and saves your progress. The request will appear in the table with a *Draft* status and can be edited at any time.
- **Submit Request:** Click **Submit Request** when all required fields are complete. The system validates all fields, generates a unique Tool Code, and routes the request to the *Pending* approval queue.

*Note: Once submitted, the request cannot be edited until it is Approved or Rejected by an Approver.*

### Editing a Draft or Rejected Request

1. On the Tool Codes table, locate your **Draft** or **Rejected** request.
2. Click the **Edit (Pencil)** icon in the Actions column.
3. If the request was **Rejected**: A red banner at the top of the form will display the Approver's exact rejection reason. Address all flagged issues before resubmitting.
4. Click **Submit Request** to send the corrected request back for approval.

Figure 4: Edit Modal — the red Rejection Reason banner guides the required corrections

## Understanding Request Statuses

| Status          | Meaning                                             |
|-----------------|-----------------------------------------------------|
| <b>DRAFT</b>    | Saved but incomplete; not yet submitted for review. |
| <b>PENDING</b>  | Submitted and awaiting an Approver decision.        |
| <b>APPROVED</b> | Finalised; Tool Code is active and immutable.       |
| <b>REJECTED</b> | Returned by an Approver with a documented reason.   |

## 5 Using the Approval Workflow

Approvers govern the accuracy of all Tool Code requests before they are finalised.

### Reviewing a Pending Request

1. Navigate to the **Tool Codes** tab. Click the **Pending** filter button to isolate requests awaiting review.
2. Click the **View (Eye) icon** on any pending row.
3. A read-only modal will open displaying all submitted data across four sections: *Tool Information*, *Specifications*, *Supplier Data*, and *User Remarks*.

**Review Tool Request**

Tool Code: DRL-SD-121X123-125-ST-T-PO

**Tool Information**

|             |               |              |                  |
|-------------|---------------|--------------|------------------|
| Description | Drilling Tool | Fixed Value  | ID — Solid Drill |
| Asset Type  | T Joint       | Raw Material | Produce Metal    |

**Drilling Tool Specifications**

|                |            |              |     |               |     |
|----------------|------------|--------------|-----|---------------|-----|
| Drill Diameter | 121        | Flute Length | 123 | No. of Flutes | 125 |
| Drill Type     | STIP DRILL |              |     |               |     |

**Supplier Information**

|            |       |                 |      |
|------------|-------|-----------------|------|
| Supplier 1 | Stock | Ordering Code 1 | 8001 |
| Supplier 2 | 185   | Ordering Code 2 | 1002 |

**User Remarks**

No remarks provided.

Recorded by: testuser on Apr 15, 2026

Approve Reject Close

Figure 5: Review Tool Request Modal — read-only data layout with action buttons

### Approving a Request

- Verify all fields are accurate and complete.
- Click **Approve**. Confirm the action in the popup dialog.



- The Tool Code status changes to *Approved* and the code becomes active. Approved codes are **immutable** and cannot be changed.

## Rejecting a Request

- Click **Reject** in the review modal.
- A secondary modal appears with a mandatory **Rejection Reason** text field.
- Enter a specific, actionable reason to guide the Requester's correction (e.g., *"Incorrect supplier selected — use Supplier Code XYZ-001"*).
- Click **Confirm Reject**. The request returns to the Requester with the documented reason displayed in their Edit modal.

## 6 Administrator Guide (Staff Only)

Users with an **Admin** or **Super Admin** role have access to Master Data management and system-wide configuration via the navigation menu.

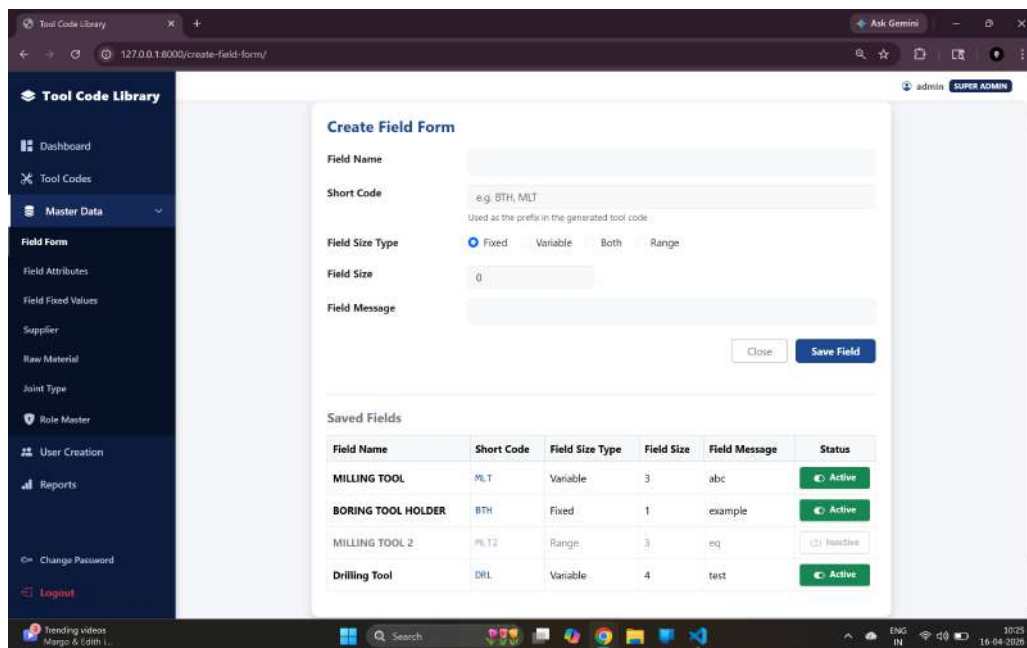


Figure 6: The Master Data Management Interface — Active vs. Inactive rows

## Master Data & Soft Deletes

All reference data (Suppliers, Materials, Joint Types) is managed through a **Soft Delete** model. Items are *never permanently deleted*. To remove an item from user-facing drop-downs:

- Locate the item in its respective Master Data table.
- Click the green **Active** toggle button.

- Confirm the status change. The row will dim and the toggle will switch to *Inactive*. The item disappears from new request forms, but all historical records that reference it remain intact.

## Field Attributes Master

This module defines the dynamic sub-fields that appear in the Create Code form when a Requester selects a specific Primary Description.

1. Navigate to **Master Data** → **Field Attributes Master**.
2. Select a **Tool Type** from the dropdown.
3. Under *Add Sub-field*, enter an **Attribute Name**, choose an **Input Type** (Number, Text, or Select), and toggle whether the field is **Required**.
4. If *Select* is chosen, the **Dropdown Builder** appears: type an option value, press **Enter** to create a pill tag, and repeat until all options are added.
5. Click **Add** to save the sub-field. Changes apply to all future requests for that tool type immediately.

**Field Attributes Master**

For each tool type, define the sub-fields that appear in the Create Code form. For **Dropdown** type attributes, add the option values before clicking Add.

Select Field / Tool Type: MILLING TOOL

Add Sub-field for: MILLING TOOL

Attribute Name: e.g. Bore Size    Input Type: Number    Order: 5    Required?: ☒    Add

Configured Sub-fields for MILLING TOOL

| Order | Attribute Name | Input Type | Required | Actions     |
|-------|----------------|------------|----------|-------------|
| 1     | Diameter       | Number     | No       | Edit Active |
| 2     | No. of Teeth   | Number     | Yes      | Edit Active |
| 3     | Overall Length | Number     | No       | Edit Active |
| 4     | Trdd           | Dropdown   | Yes      | Edit Active |

Figure 7: Field Attributes Master

## Role Master

Roles define the exact combination of tab access and workflow permissions. Once a role is created here, it becomes available to assign to individuals in the **User Creation** tab.

1. Go to **Master Data** → **Role Master** and click + **Create New Role**.
2. Enter a descriptive Role Name (e.g., *Senior Approver*).

3. Use the toggle switches to grant permissions such as *Can Approve Requests* and *Access Reports Tab*.
4. Click **Create Role** to save. This role will now appear in the Role dropdown when creating a new user.

*Note: If you edit the permissions of an existing role, those changes will take effect immediately for all users currently assigned to that role.*

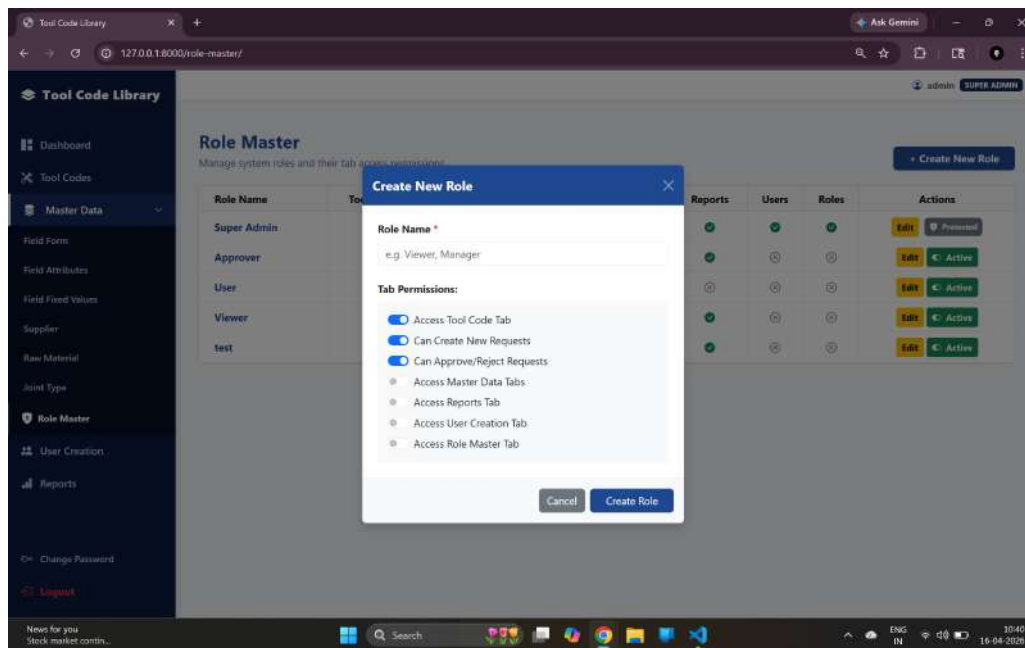


Figure 8: Role Master

## Master Reporting

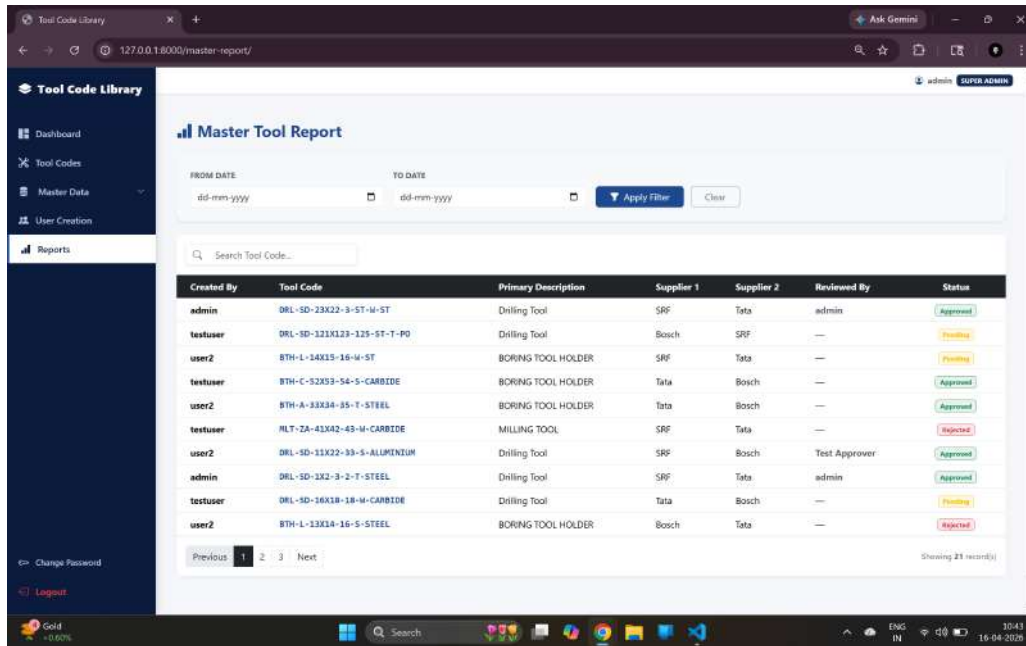


Figure 9: Master Report Interface — Date filters, Search Bar, and paginated table

- Navigate to the **Reports** tab in the sidebar.
- Select a **From Date** and **To Date** to scope the dataset.
- Use the **Dynamic Search Bar** to filter instantly by Tool Code without reloading the page.
- Use the **Pagination Controls** at the bottom to navigate large result sets.

## 7 Troubleshooting & FAQs

**Q: I selected the same supplier in both Supplier 1 and Supplier 2, but the system threw a validation error.**

A: Supplier 1 and Supplier 2 must always be different selections. Choose a distinct supplier for each field before submitting.

**Q: My request is marked as Rejected. What should I do?**

A: Click the Edit (Pencil) icon on the rejected row. A red banner at the top of the form will display the Approver's exact rejection reason. Address the flagged issues and click Submit Request to resubmit.

**Q: Can I edit an Approved Tool Code?**

A: No. Approved Tool Codes are immutable. To update a tool specification, you must submit a new Tool Code request with the correct information.

**Q: A dropdown field is empty when I try to create a request.**

A: The Master Data item linked to that dropdown may have been deactivated by an

Administrator. Contact your Administrator to reactivate the relevant Supplier, Material, or Joint Type entry.

**Q: My request was submitted but I cannot find it in the table.**

A: Use the status filter buttons (All, Pending, Draft, etc.) to ensure the correct view is selected. You may also use the search bar to locate the request by its tool code.

**Q: If I edit a draft request, does the Tool Code change?**

A: Drafts are assigned a temporary placeholder [DRAFT]. The actual unique Tool Code is only generated the moment you successfully submit the request for approval. Editing a draft simply updates your input data before that final generation.

**Q: I forgot my password and cannot log in.**

A: Passwords cannot be self-recovered. Contact a System Administrator, who can locate your account in the User Management section and set a new temporary password for you.

**Q: How are Tool Codes generated?**

A: The system assembles the code automatically from the structured attribute inputs you provide during request creation, producing a standardised, sequential, and unique identifier of up to 40 characters.